

Q3

Magnification = image size / actual size

= 40 mm / 0.08 mm

= 500 times

Q1 (a)

Photosynthesis is the process by which green plants make glucose from carbon dioxide and water using light energy.

Q1 (b)

carbon dioxide + water → glucose + oxygen

Q5

Aerobic respiration uses oxygen and releases much more energy.

Anaerobic respiration does not use oxygen and releases less energy. In muscles it produces lactic acid, not carbon dioxide.

Q2

Stomata are tiny pores, mostly on the lower leaf surface.

Carbon dioxide diffuses into the leaf through them, while oxygen and water vapour diffuse out. Guard cells change shape to open or close each pore depending on water and light conditions...

Q2 continued

When guard cells take in water they become turgid and curve apart, opening the stoma. When they lose water they become flaccid and the pore closes. This controls gas exchange and also limits water loss by transpiration.

Q6

Mitosis produces two genetically identical daughter cells.

It is used for growth and repair of tissues.

I did not answer the graph question.